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Integrated hydrochemical and microbiological assessment of groundwater quality for early contamination detection and preventive management

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Abstract

Groundwater is a critical source of drinking water in many rural areas and peri-urban communities; however, both geochemical and microbiological factors increasingly influence its suitability. This study presents an integrated assessment of groundwater quality in mining and agriculture-influenced communities of Ghana (Segyemase, Juaso and Nsuapemso) using hydrochemical characterization, water quality indices, microbiological indicators and risk-based interpretation. Sixteen borehole water samples were analysed for physicochemical parameters, major ions and selected trace metals, alongside microbial indicators including total coliforms, faecal coliforms, *Escherichia coli*, *Salmonella spp.* and *Vibrio spp.* Hydrogeochemical processes were evaluated using ionic ratios, saturation indices and Piper diagrams, while groundwater suitability for drinking was assessed using the Water Quality Index (WQI) and complementary pollution indices. Microbiological quality was evaluated through descriptive statistics, exceedance analysis and risk-based assessment. Results indicate that groundwater chemistry is predominantly controlled by the water-rock interaction process, with Ca-HCO₃ and Ca-Mg-SO₄ facies reflecting spatial variability linked to lithology and mining influence. All sampled boreholes were classified as excellent for drinking based on WQI values. Pollution indices (GPI, WPI and CPI) generally indicated low to moderate physicochemical impairment, whereas the Water Nutrient Pollution Index (WNPI) identified localised nutrient impacts and the Ecological Risk Index (ERI) classified all samples as low ecological risk. However, microbiological assessment revealed the widespread occurrence of total coliforms and the localized detection of faecal coliforms and *E. coli* in two boreholes, indicating sanitary vulnerability and a potential public health risk. These findings demonstrate that chemical suitability does not necessarily equate to microbiological safety. Integrated interpretation enabled prioritization of high-risk boreholes for targeted intervention and informed community-level risk communication. The study highlights the importance of combining hydrochemical assessment, water quality indices, pollution indices, microbiological monitoring and risk-based frameworks to support



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early contamination detection, evidence-based groundwater management and preventive public health protection.

Keywords Groundwater quality, Hydrochemical indicators, Pollution indices, Microbiological contamination, Drinking water safety, Preventive resource management, Ghana

1 Introduction

Groundwater is increasingly becoming an important source of freshwater due to pressure on surface water systems, safety and the persistence of inadequate supply of treated water [1, 2]. Though many regions in Ghana are endowed with streams, rivers, lakes and wetlands, human activities have significantly compromised the quality of these surface resources. In recent years, Ghana has experienced surface water degradation associated with mining activities, agricultural runoff, improper waste disposal and sedimentation, which has reduced the quality of these water sources for domestic use [3]. Consequently, groundwater has become the most reliable alternative source of potable water, not only for rural areas but also for peri-urban and urban areas where pipe water supply remains insufficient to meet the growing demand [4]. Even in major cities, including Accra, households supplement an unreliable supply of municipal water with boreholes and hand-dug wells [5].

The quality of groundwater is influenced by an interplay of both natural and anthropogenic factors, including the geological and hydrological characteristics of the aquifer, the mineralogical composition of the host rocks, recharge conditions, soil properties, and the nature and intensity of land use within the catchment area [6]. Generally, in hard rock terrains, the occurrence of groundwater is confined to weathered zones and fractures within the bedrock, making water quality sensitive to interactions between water and rock, as well as surface-derived inputs [7]. Chemical processes such as mineral dissolution, ion exchange and redox reaction tend to control the natural evolution of groundwater chemistry [8]. Anthropogenic activities such as mining and intensive agriculture introduce foreign constituents that alter this balance and degrade water quality [9, 10].

When evaluating the suitability of groundwater for domestic and other uses, it is essential to understand its hydrogeological characteristics [8, 11]. Hydrogeochemical investigations provide insights into the processes governing groundwater composition, origin of dissolved ions and anthropogenic influences, or the extent to which water quality reflects natural conditions [12, 13]. These assessments are extremely important for areas undergoing rapid land-use change, where intensive mining and agricultural activities accelerate the movement of nutrients, trace elements and dissolved solids into aquifer systems [14]. In the absence of systematic evaluation, the deterioration of groundwater quality may not be detected until it causes harm to human health and water safety.

Integrated approaches to groundwater quality evaluations have proven effective in providing comprehensive hydrogeochemical data for holistic information, meaningful assessment and decision-making [15, 16]. Water quality and pollution indices generally provide simple but comprehensive means of assessing groundwater suitability by combining multiple physicochemical parameters into unified interpretive values [17]. Hydrogeochemical characterisation and statistical analysis, when used together, serve as tools that aid in differentiating natural geochemical controls from contamination arising from

anthropogenic activities, also enabling spatial comparison of groundwater quality across different locations [18].

Surface water resources naturally occur in the study area and ongoing anthropogenic activities, particularly mining in water bodies and the widespread use of agrochemicals, have greatly reduced the quality and reliability of surface water [19]. As a result, groundwater has become the main source of water for households in rural settlements and growing urban communities in the study area. The rising dependence on groundwater provides a basis for a more detailed assessment of its hydrogeochemical characteristics and suitability for human use [20].

Despite these efforts, integrated regional-scale groundwater investigations that simultaneously evaluate hydrogeochemical processes, multiple pollution diagnostics, ecological risk indicators and microbiological contamination remain limited in the Eastern Region of Ghana.

Previous groundwater-quality studies in eastern Ghana and adjacent mining-agricultural terrains have largely focused on hydrochemical characterisation, trace-metal contamination, and drinking-water suitability using conventional indices and multivariate statistical tools [21]. In parallel, national and local institutions, including the Water Resources Commission (WRC), the Ghana Water Company Limited (GWCL), the Community Water and Sanitation Agency (CWSA) and Environmental Health Units of District Assemblies, routinely conduct borehole surveillance and sanitary inspections to support water-supply planning and public-health protection [22]. However, such monitoring programmes typically emphasize physicochemical compliance testing, with comparatively few assessments integrating microbiological indicators, ecological risk diagnostics and multi-index pollution frameworks within a single management-oriented evaluation [8].

In this context, the present study extends existing groundwater research and monitoring efforts in the area and other parts of the world, by combining hydrogeochemical process interpretation with a comprehensive suite of water-quality and pollution indices (WQI, GPI, CPI, TMPI, WNPI, and ERI), microbiological indicators, and risk-based prioritization of boreholes. Unlike earlier investigations that considered chemical or microbial aspects in isolation, this work adopts an integrated chemical-microbiological-ecological approach to identify early warning signals of contamination, distinguish geogenic from anthropogenic controls and generate actionable evidence to support targeted remediation, routine surveillance and sustainable groundwater management in a region where dependence on subsurface water resources is rapidly increasing.

This study aims to characterise the physicochemical and hydrogeochemical properties of groundwater in the area, identify the major processes governing groundwater chemistry and evaluate groundwater quality using integrated assessment indices. Through a meaningful baseline of groundwater conditions, the study will support early detection of quality deterioration, guide targeted remediation efforts for areas where contamination is detected and inform subsequent surveillance and sustainable management of groundwater in a region where reliance on subsurface water resources is intensified.

2 Study area description

2.1 Location

The samples were collected from three communities: Segyemase in the Abuakwa South municipality and Juaso and Nsuapemso in the Fanteakwa South District of the Eastern Region of Ghana (Fig. 1). The two neighbouring administrative districts are located between latitudes $6^{\circ} 02' 00.0''$ and $6^{\circ} 18' 50.4''$ N and longitudes $0^{\circ} 29' 0.0''$ W and $0^{\circ} 35' 34.0''$ W. Kibi and Osino are the capitals of the administrative districts, approximately 100 km from Accra. The three communities have a long history of surface mining. Cocoa, coconut, and oil palm are some of the cash crops cultivated in the districts, alongside other crops such as cassava, plantain, and vegetables. Other economic activities in the study area include trading, tailoring, hairdressing, etc.

2.2 Physiography, climate and vegetation

The study area is predominantly undulating, characterised by hills and valleys typical of the Birimian Proterozoic basement complex [23]. Soil is generally sandy to loamy with variation in between, influenced by the nature of the topography and drainage patterns of the area. Landscape and underlying geology strongly influence groundwater recharge, flow and chemistry.

The region has a tropical climate, with a wet season from April to October and a dry season from November to March. Average temperatures range from 24 to 30 °C and the annual mean rainfall of 1750 mm supports the vegetation [23]. Variations in the seasonal rainfall and temperature greatly affect the rate of groundwater recharge and dilution of dissolved ions.

Vegetation ranges from cultivated farmland (mostly cash crops) to secondary forests [24]. The Fanteakwa South District has more intensive anthropogenic activities,

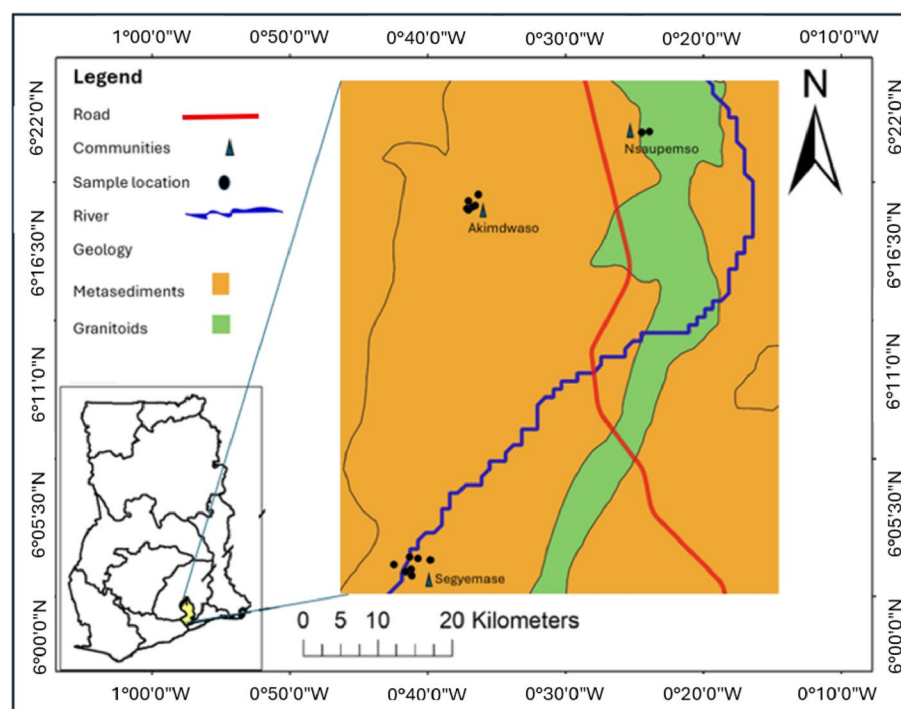


Fig. 1 Location and geological map of the study area

particularly agricultural production and mining, compared to Abuakwa South Municipality [24]. The vegetation cover and land use change can influence groundwater chemistry with respect to trace metal concentrations and total dissolved solids [6].

2.3 Geology and hydrogeology

The study area forms part of the northern Kibi-Winneba belt, one of the six Proterozoic metavolcanic greenstone belts in Ghana [23]. The underlying geology of the area is characterised by metavolcanic and metasedimentary rocks intruded by granitic and gneissic bodies. These lithological units are generally crystalline, hard, with low primary porosity, a typical feature of the Birimian geological terrain [25, 26].

Groundwater occurrence within the area is therefore dominantly controlled by secondary porosity developed through weathering, fracturing and jointing of the basement rocks. Similar to other Birimian terrains in Ghana, groundwater is mainly hosted in the weathered regolith and fractured bedrock aquifers, which constitute the principal water-bearing zones [20, 26–29].

The hydrogeology of the Birimian Province is structurally controlled with fractures, joints, faults and shear zones acting as preferential conduits and storage for groundwater [20, 26]. These geological discontinuities enhance the transmissivity of groundwater, largely determining well yields and groundwater flow patterns across the basement.

Groundwater system recharge mainly through the direct infiltration of rainwater, facilitated by seasonal precipitation and tropical climatic conditions [30]. Recharge rates and movement are further influenced by the degree of weathering and connectivity of the fracture networks within the basement rocks [20].

Groundwater occurrence in the area is primarily controlled by secondary porosity developed within the weathered regolith and fractured bedrock typical of Precambrian basement terrains [27]. The aquifer system is generally unconfined to semi-confined, with saturated weathered zones ranging from approximately 10 to 30 m in thickness, underlain by fractured crystalline rocks that form the main productive horizons [26]. Static water levels in boreholes typically occur at depths of about 5 to 10 m below ground level, reflecting seasonal recharge conditions [20]. Reported borehole yields commonly range between 1 and 5 m³h⁻¹, although higher yields are locally recorded where fracture density is enhanced along structural zones and valley bottoms [26, 31].

In general, the occurrence, flow and storage of groundwater within the study area reflects the combined influence of Birimian geology, weathering intensity and structural controls, consistent with national-hydrogeological models for crystalline basement terrains in Ghana [32]. This framework of geology and hydrology provides an appropriate basis for interpreting the observed groundwater hydrochemistry, pollution indices and multivariate statistical patterns identified in the study.

3 Materials and methods

3.1 Sampling and analysis

3.1.1 Sampling

Groundwater sampling was designed to capture the main drinking water sources used by the residents in the study communities. All operational boreholes serving domestic purposes within the selected communities at the time of fieldwork were sampled, rather than a subset, resulting in a total of sixteen sampling points. This census-type approach

reflects the boreholes most relied upon for household water supply and therefore provides a realistic screening of potential exposure pathways and contamination risks at the community scale. Although the number of boreholes differs among communities owing to variations in infrastructure availability, the coverage represents the full set of functional groundwater sources present during the survey.

Sampling was conducted in September 2022, during the late wet season, when recharge is active and dilution or mobilization effects may influence groundwater chemistry. The seasonal context is therefore noted and future investigations are recommended to include multi-seasonal monitoring to evaluate temporal variability.

To assess sampling and analytical precision, one field duplicate sample was collected from a selected borehole in each of the three study communities, resulting in a total of three field duplicate samples. The duplicate water samples were drawn consecutively from the same borehole following purging and stabilization of field parameters and were treated identically to the primary samples during preservation, transport and laboratory analysis. For physicochemical and major ion analysis, samples were collected in pre-cleaned polyethylene bottles; those intended for cation and trace metal determination were acidified onsite to $\text{pH} < 2$ using ultrapure nitric acid, whereas anion samples were left unacidified and stored cool before analysis.

Samples for bacteriological analysis were collected separately in sterile 500 mL polypropylene bottles containing sodium thiosulphate to neutralize residual disinfectants where present. Borehole outlets were disinfected with ethanol (70%) and flamed before sampling and water was allowed to flow for several minutes to avoid stagnant water in the pipes. Bottles were opened only at the time of collection and filled without touching the rim or interior surfaces. All microbiological samples were immediately placed in insulated cool boxes with ice packs and transported to the laboratory at approximately 4 °C. Analyses were initiated within 5–12 h of collection in accordance with standard methods. During transport and storage, samples were protected from light and physical disturbance to minimize microbial alteration before analysis. Each sample was assigned a unique sample code for every sample location. Sample collection, preservation, and transportation to the laboratory followed the guidelines recommended by the American Public Health Association [1].

3.1.2 Laboratory analysis

The sampled groundwater was analysed to determine its suitability for domestic use and public health by measuring physicochemical, nutrient, trace element and bacteriological parameters using the recommended standard procedures of the American Public Health Association [33] and related guidelines at the laboratories of Council for Scientific and Industrial Research- Water Research Institute (CSIR-WRI) in Accra, Ghana.

Electrical conductivity (EC) and pH were measured in situ using a calibrated portable multiparameter meter, while temperature was recorded using a digital thermometer. Colorimetric and nephelometric methods were used to determine colour and turbidity, respectively. Total suspended solids (TSS) were measured by filtration followed by oven-drying, while Total dissolved solids (TDS) were estimated gravimetrically [33].

Major ions include bicarbonate (HCO_3^-), calcium (Ca^{2+}), carbonates (CO_3^{2-}), Chloride (Cl^-), and magnesium (Mg^{2+}), which were analysed by titration. Total hardness, calcium hardness and magnesium hardness were determined by EDTA titration. Total

alkalinity was measured by strong acid titration expressed as CaCO_3 [33]. Phenolphthalein alkalinity was not detected in any of the samples. Similarly, carbonates (CO_3^{2-}) alkalinity and hydroxide (OH^-) species were not detected. Therefore, total alkalinity was entirely attributable to bicarbonate (HCO_3^-).

Concentrations of potassium (K^+) and sodium (Na^+) were determined by using flame photometry. Sulphate (SO_4^{2-}) was analysed using spectrophotometric methods. Fluoride concentrations were measured with the SPADNS colorimetric method [34]. Nutrient parameters, nitrate-nitrogen (NO_3^- -N), nitrite-nitrogen (NO_2^- -N), phosphate phosphorus (PO_4^{3-} -P) and ammonium-nitrogen (NH_4^+ -N), were determined by spectrophotometry following the protocols of the American Public Health Association [33].

Concentration of trace elements was determined at CSIR-WRI, Accra, Ghana, using an Agilent 4210 Microwave Plasma-Atomic Emission Spectrophotometry (MP-AES; Aligent Technologies, USA) after filtration and acidification of samples to $\text{pH} < 2$ with ultra-trace metal grade pure nitric acid (HNO_3 , Merck, Germany). To ensure analytical accuracy, certified reference standards, procedural blanks and duplicate measurements were used in calibration.

Microbiological analysis was conducted using the membrane filtration technique. The water samples were filtered through sterile $0.45\ \mu\text{m}$ pore-size membrane filters (47 mm diameter), after which the filters were aseptically transferred onto selective culture media for bacterial enumeration. Total coliforms (TC), faecal coliform (FC) and *Escherichia coli* (*E. coli*) were cultured on Chromocult Coliform (Merck, Germany), while *Salmonella spp.* and *Vibrio spp.* were isolated using Xylose Lysine Deoxycholate (XLD) agar and Thiosulfate Citrate Bile Salts Sucrose (TCBS) agar, respectively (Oxoid Ltd., UK). Plates were incubated under appropriate temperature and time conditions before colony enumeration. The results from the analysis were expressed in colony-forming units per 100 mL (CFU/100 mL).

Charge balance error was calculated for major ions, following the method of Hounslow [35], to ensure data reliability before further processing and interpretation. The acceptable charge balance error was strictly within $\pm 10\%$, using the mathematical expression:

$$\text{CBE (\%)} = \frac{\sum \text{Cations} - \sum \text{Anions}}{\sum \text{Cations} + \sum \text{Anions}} \times 100 \quad (1)$$

All the ionic concentrations were expressed in milliequivalents per litre (Meq/L). Where $\sum \text{Cations}$ is the sum of all cations and $\sum \text{Anions}$ is the sum of all the anions.

3.2 Data analysis and statistical processing

The physicochemical, trace element and bacteriological characteristics of the samples were first summarized with descriptive statistics. The dataset was assessed for suitability for multivariate analysis using the Kaiser-Meyer-Olkin (KMO) measure of sampling adequacy and Bartlett's test of sphericity to confirm that the variables were sufficiently interrelated for factor analysis [36]. The KMO value of 0.585 exceeded the recommended threshold of 0.50, indicating acceptable sampling adequacy, while Bartlett's Test of Sphericity was statistically significant ($\chi^2 = 280.573$, $\text{df} = 45$, $p < 0.001$), confirming the suitability of the dataset for dimensionality reduction through PCA.

Before correlation and multivariate analyses, the distribution of the data was assessed using the Shapiro-Wilk normality test. The test indicated that most hydrochemical

variables deviated from normal distribution at $p < 0.05$, which justified the use of standardized data transformation and multivariate statistical approaches for hydrochemical interpretation.

Spearman's rank correlation analysis was employed to determine the relationship between the measured groundwater quality parameters. Spearman correlation was chosen due to its robustness to non-normal data distribution and outliers, which are common in hydrogeochemical and environmental datasets. The correlation matrix was used to identify associations among major ions, trace elements, providing preliminary insight into possible common sources and controlling processes. Correlation analysis identifies pairwise relationships; however, it does not define clusters or groupings within the overall dataset.

Factor analysis (FA) with principal component extraction was therefore applied to identify the factors controlling groundwater chemistry in the study area and group correlated variables into components. Varimax rotation was used to maximize interpretability of the data and reduce cross-loadings, and the factors extracted represent dominant hydrogeological and environmental processes.

The integrated approach provided a strong framework for identifying natural against human-induced influences on groundwater quality, which was used in the interpretation of complex hydrogeochemical datasets.

3.3 Hydrogeological and water quality assessment

3.3.1 Hydrogeological facies and geochemical controls

Piper trilinear diagram, Gibbs diagrams and bivariate diagrams were generated using AquaChem 5.0. Major ions were converted from mg/L to Meq/L before plotting. The Piper diagram was used to classify groundwater into dominant hydrogeochemical facies based on the relative proportions of major cations (Ca^{2+} , Mg^{2+} , Na^+ , K^+) and anions ($\text{HCO}_3^- + \text{CO}_3^{2-}$, SO_4^{2-} , Cl^-) [37, 38]. This provided insight into groundwater evolution and the dominant geochemical controlling process [39]. The Gibbs diagram was used to identify the principal mechanisms controlling groundwater chemistry by plotting TDS against the ratios $\text{Na}^+ / (\text{Na}^+ + \text{Ca}^{2+})$ and $\text{Cl}^- / (\text{Cl}^- + \text{HCO}_3^-)$ [13]. These plots helped to distinguish between rock-interaction, precipitation dominance and evaporation-crystallization processes. Bivariate plots from the selected ionic ratios of $\text{Na}^+ / \text{Cl}^-$, $\text{Ca}^{2+} / \text{Mg}^{2+}$, $(\text{HCO}_3^- + \text{CO}_3^{2-}) / (\text{Ca}^{2+} + \text{Mg}^{2+})$ and $\text{Cl}^- / (\text{HCO}_3^-)$ were further employed to infer contributions of silicate weathering, carbonate dissolution, ion exchange and possible anthropogenic effects on the composition of groundwater [35].

3.3.2 Saturation index

The saturation index (SI) was calculated for all samples to assess the equilibrium state of the groundwater, with the selected mineral phases being calcite, dolomite, gypsum, halite, and fluorite. The saturation index was calculated with the expression:

Where IAP is the ion activity product of the dissolved mineral species, and K_{sp} is the solubility product constant of the mineral at the measured temperature.

A SI value less than zero indicates undersaturation (mineral dissolution), $\text{SI} \approx 0$ indicates equilibrium and greater than zero indicates supersaturation (mineral precipitation) [40].

3.3.3 Pollution indices

Guideline concentrations (S_i) used in the computation of the indices were adopted from the World Health Organization (WHO) drinking water quality standard [41]. Multiple pollution indices were applied to capture different dimensions of groundwater quality. The General Pollution Index (GPI), Comprehensive Pollution Index (CPI) provide overall contamination status, whereas the Water Nutrient Pollution Index (WNPI) emphasizes maximum parameter exceedance. The trace Metal Pollution Index (TMPI) specially evaluates heavy metal accumulation, while Ecological Risk Index (ERI) assesses potential ecological and toxicological impacts. The combined use of these indices allows for a multi-dimensional evaluation rather than reliance on a single metric.

3.3.3.1 Groundwater Pollution Index The Groundwater Pollution Index was calculated to assess the overall pollution status of groundwater. It was calculated by integrating selected physicochemical and trace element parameters [21, 42, 43]. GPI was computed as:

$$GPI = \frac{1}{n} \sum_{i=1}^n \frac{C_i}{S_i} \quad (3)$$

There C_i is the measured concentration (mg/L) of parameter i , S_i is the corresponding guideline value (mg/L) and n is the number of parameters considered. GPI values less than 1 indicate unpolluted groundwater, whereas values greater than 1 indicate increasing levels of pollution [43].

3.3.3.2 Comprehensive Pollution Index (CPI) The Comprehensive Pollution Index was used to estimate the cumulative pollution effect of multiple parameters. CPI was determined using the equation:

$$CPI = \frac{1}{n} \sum_{i=1}^n P_i \quad (4)$$

Where

$$P_i = \frac{C_i}{S_i} \quad (5)$$

CPI values were classified to reflect different pollution categories, ranging from clean to heavily polluted groundwater [21, 44]. Comprehensive Pollution Index (CPI) was classified ≤ 0.20 as Clean, 0.21–0.40 as slight pollution, 0.41–1.00 as moderate pollution and > 1.00 as heavy pollution.

3.3.3.3 Water Nutrient Pollution Index (WNPI) The water Nutrient Pollution index was calculated to assess groundwater contamination due to nutrients, particularly NO_3^- -N, NO_2^- -N, NH_3^- -N and PO_4^{3-} -P. WNPI was computed as:

$$NPI = \frac{1}{n} \sum_{i=1}^n \frac{C_i}{S_i} \quad (6)$$

where C_i is the measured concentration of nutrient I (mg/L), S_i is the corresponding guideline value (mg/L) and n is the number of nutrient parameters [44, 45]. Higher WNPI values indicate increased nutrient enrichment and potential anthropogenic influence [41, 46–49]. Water Nutrient Pollution Index was classified as: ≤ 0.10 was unpolluted, 0.11–0.20 as slight nutrient pollution, 0.21–0.50 as moderate nutrient pollution as > 0.50 as severe nutrient pollution.

3.3.3.4 Trace Metals Pollution Index (TMPI) The Trace Metals Pollution Index was applied to evaluate groundwater contamination from trace metals. TMPI was calculated as:

$$TMPI = \frac{1}{n} \sum_{i=1}^n \frac{C_i}{S_i} \quad (7)$$

Where C_i is the measured concentration of metal I, S_i is the corresponding guideline value and n is the number of trace metals analysed. TMPI Values increase with increasing metal contamination. Trace Metal Pollution index (TMPI) was classified as: ≤ 0.10 Low, 0.11–1.00 Moderate, 1.01–2.00 High > 2.00 , Very high [21, 50].

3.3.3.5 Ecological Risk Index (ERI) The Ecological Risk Index was employed to evaluate the potential ecological risk posed by toxic trace metals in groundwater. ERI was calculated following:

$$ERI = \sum_{i=1}^n E_i^r \quad (8)$$

where

$$E_r^i = T_r^i \times \frac{C_i}{S_i} \quad (9)$$

E_r^i represents the potential ecological risk factor for metal i , T_r^i is the toxic response factor for metal i , C_i is the measured concentration (mg/L), and S_i is the reference of the guideline value (mg/L). ERI values were used to classify groundwater into four categories: low, moderate, considerable, and high ecological risk [51]. Following Hakanson [51], the potential ecological risk index (ERI) was classified as: < 40 Low risk, 40–80 Moderate risk, 80–160 Considerable risk, 160–320 High risk, > 320 Very high risk.

3.3.4 Groundwater suitability for drinking

The borehole samples were evaluated for their suitability as drinking water using the Water Quality Index (WQI) method [21, 43, 44, 52], which integrates multiple physicochemical and chemical parameters into one numerical value reflecting the general potability of the water [41, 42, 53]. The parameters were assigned relative weights based on their significance to human health and the quality of drinking water. Although the weighted arithmetic WQI approach involves some subjectivity in assigning parameter weights, it remains widely applied in groundwater quality studies. It allows comparison with previous work in similar hydrogeological settings. This limitation is acknowledged

here and further studies may explore entropy-weighted WQI (EWQI) approaches for improved objectivity. The relative weights (W_i) were calculated with the equation:

$$W_i = \frac{w_i}{\sum_{i=1}^n w_i} \quad (10)$$

where W_i = assigned weight of parameter i , n = total number of parameters.

For each parameter, the quality rating (q_i) was computed as:

$$q_i = \frac{C_i}{S_i} \times 100 \quad (11)$$

Where C_i = measured concentration of the parameter i (mg/L), S_i = drinking water guideline value of the parameter (mg/L) [41].

For pH, since the WHO provides a permissible range (6.5–8.5) rather than a single standard value, a modified quality rating equation was applied. The ideal value ($V(\text{ideal})$) was taken as 7.0 and the permissible limit (S_i) was taken as 8.5. The quality rating for pH was therefore calculated using:

$$\frac{(V_{pH} - 7.0)}{(8.5 - 7.0)} \times 100 \quad (12)$$

Then the sub-index (SI) for each parameter was obtained using the equation:

$$SI = w_i \times q_i \quad (13)$$

Finally, the overall Water Quality Index was given by the equation:

$$WQI = \sum_{i=1}^n SI_i \quad (14)$$

The WQI was interpreted based on the classification below:

WQI < 50 Excellent, 50–100 Good, 100–200 Poor, 200–300 Very Poor, > 300 Unsuitable.

3.4 Microbial analysis and risk assessment

3.4.1 Microbial analysis

Total coliform, faecal coliforms and *Escherichia coli* were enumerated using the membrane filtration technique following Standard Methods for the Examination of Water and Wastewater [1, 33]. Aliquots of 100 mL were filtered through sterile 0.45 μm membrane filters, which were then placed on selective culture media. m-Endo agar was used for total coliforms and incubated at 37 °C for 24 h, while m-FC agar was used for faecal coliforms and incubated at 44.5 °C for 24 h. *E. coli* was confirmed using chromogenic media. Colony counts were expressed as colony-forming units per 100 mL (CFU/100 mL) [1]. Procedural blanks and duplicate analyses were included for quality control.

3.4.2 Risk assessment

Percent exceedance was calculated to quantify how many samples exceeded WHO drinking guidelines [41] (0 CFU/100mL for TC, FC, and *E. Coli*), the following formula was used:

$$\text{Percentage Exceedance} = \frac{\text{Number of samples above guideline}}{\text{Total number of samples}} \times 100 \quad (15)$$

The Microbial Contamination Index (MCI) was calculated to standardize microbial counts and prioritize boreholes for intervention [54]

$$MCI = \frac{\text{Measured Count (CFU/100mL)}}{\text{WHO Guideline (CFU/100mL)} + 1} \quad (16)$$

Adding 1 to the denominator avoids division by zero and allows direct ranking of contamination severity [54].

Microbial risk levels of the samples were classified based on TC and *E. coli* count [15] as follows: low risk when no organisms were detected (0 CFU/100 mL), moderate risk for concentrations between 1 and 10 CFU/100 mL and high risk for values exceeding 10 CFU/100 mL.

The probability of infection was estimated using standard QMRA exposure assumptions and did not explicitly stratify populations by gender or body type. A representative adult exposure scenario was adopted based on typical drinking water ingestion rates reported in the literature. Age-specific scenarios (e.g., children) were not separately evaluated and are acknowledged as a limitation of this assessment.

The annual probability of infection (P_{annual}) due to ingestion of *E. coli* contaminated groundwater was estimated using a Beta-Poisson dose-response model [55]:

$$P_{\text{inf}} = 1 - \left(1 + \frac{D}{\beta}\right)^{-\alpha} \quad (17)$$

Where: P_{inf} = probability of infection per exposure.

D = microbial dose per day (CFU/day).

α = 0.1705 (Pathogens dose per day (CFU/day).

β = 1.61×10^6 (CFU, pathogen-specific).

The daily microbial dose was calculated as:

$$D = C \times V \quad (18)$$

where C = *E. coli* concentration in water (CFU/100 ml).

V = daily ingestion volume (2 L/day, or 2000 mL/day).

The annual probability of infection was calculated using:

$$P_{\text{annual}} = 1 - (1 - P_{\text{inf}})^{365} \quad (19)$$

4 Results and discussion

4.1 Water chemistry

4.1.1 Hydrochemical characterisation

The summary statistics of the physicochemical parameters (Table 1) provide a first-order constraint on the hydrochemical regime of the study area and form the basis for interpreting dominant geochemical processes, water–rock interaction, and potential quality concerns. Groundwater pH values range from 5.38 to 7.29, with a mean of 6.49, indicating dominantly slightly acidic to near-neutral conditions. This pH distribution suggests limited buffering capacity in parts of the aquifer system, consistent with groundwater circulation through silicate-dominated lithologies and weathered regolith units that are

Table 1 Summary statistics of the analysed physicochemical parameters from the study area

Sample ID	Minimum	Maximum	Mean	Standard deviation	WHO Limit (2017)	% Sample exceeding limit
pH	5.38	7.29	6.49	0.57	6.5–8.5	43.8
Conductivity ($\mu\text{S}/\text{cm}$)	37.30	373.40	173.48	107.09	1000	0
Turbidity (NTU)	13.20	139.00	49.83	46.39	5	31.3
TSS (mg/L)	11.00	119.00	43.50	40.25	500	0
TDS (mg/L)	24.95	312.80	135.37	91.66	1000	0
ALK (mg/L)	8.40	176.20	67.66	52.70	200	0
HCO_3^- (mg/L)	10.25	214.96	82.54	64.29	250	0
T. Hard. (mg/L)	13.20	174.40	71.82	50.56	500	0
Ca^{2+} (mg/L)	0.80	52.58	17.32	16.24	75	0
Mg^{2+} (mg/L)	2.04	12.51	6.93	3.29	50	0
Cl^- (mg/L)	2.08	28.29	9.40	6.61	250	0
Na^+ (mg/L)	3.00	13.60	7.85	3.11	200	0
K^+ (mg/L)	0.30	1.80	0.67	0.37	12	0
NO_3^- (mg/L)	0.01	4.25	0.89	1.42	50	0
NO_2^- (mg/L)	0.01	0.04	0.02	0.01	3	0
PO_4^{3-} (mg/L)	0.03	0.25	0.11	0.07	5	0
SO_4^{2-} (mg/L)	0.50	25.11	10.11	7.69	250	0
F^- (mg/L)	0.00	0.29	0.17	0.09	1.5	0
Fe (mg/L)	0.01	5.94	2.25	2.29	0.3	43.8
Mn (mg/L)	0.01	1.18	0.30	0.35	0.1	43.8
Cu (mg/L)	0.01	0.11	0.03	0.04	2	0
Zn (mg/L)	0.01	0.11	0.03	0.03	3	0

widespread in many Ghanaian hydrogeological settings. Slightly acidic groundwater is commonly attributed to CO_2 -rich recharge waters, organic matter decomposition within the soil zone, and the absence or scarcity of carbonate minerals capable of providing strong alkalinity buffering [56]. From a water quality perspective, pH values below 6.5 may increase corrosivity and enhance the mobilisation of redox-sensitive metals such as iron and manganese, which is reflected in their elevated concentrations in the dataset [57].

Electrical conductivity (37.3–373.4 $\mu\text{S}/\text{cm}$; mean 173.5 $\mu\text{S}/\text{cm}$) and total dissolved solids (24.95–312.8 mg/L; mean 135.4 mg/L) indicate low to moderate mineralisation and generally fresh groundwater. These values are characteristic of aquifers with relatively short residence times and active recharge, where solute acquisition is controlled primarily by early-stage water–rock interaction rather than prolonged evaporative concentration or saline intrusion. Similar ranges have been widely reported for basement complex and weathered aquifer systems across Ghana, where groundwater chemistry is largely governed by silicate weathering reactions and limited ion accumulation [27, 58]. The observed EC–TDS relationship further supports the absence of widespread salinity stress, implying that dissolved solids are unlikely to pose direct health concerns under current conditions [41]. In contrast, turbidity (13.2–139 NTU; mean 49.83 NTU) and total suspended solids (11–119 mg/L; mean 43.50 mg/L) are notably elevated for groundwater. Such values point to significant particulate loading, which is typical of well-protected, deep groundwater systems. High turbidity and TSS in groundwater commonly reflect shallow well construction, inadequate casing or sealing, pumping-induced disturbance of fine-grained regolith materials, or direct ingress of surface runoff [57]. While turbidity itself is not a toxic parameter, its presence degrades aesthetic quality. It

may increase microbial risk by shielding pathogens from disinfection processes, an issue of particular relevance in rural and peri-urban Ghanaian water supplies.

Alkalinity (8.4–176.2 mg/L; mean 67.66 mg/L) and bicarbonate concentrations (10.25–214.96 mg/L; mean 82.54 mg/L) indicate that the carbonate system plays a moderate but variable role in controlling groundwater chemistry. The coexistence of modest alkalinity with slightly acidic pH suggests that bicarbonate is derived largely from CO₂-driven silicate weathering and soil-zone processes rather than extensive dissolution of carbonate minerals. This geochemical signature is consistent with aquifers developed within crystalline basement rocks and weathered profiles, where bicarbonate acts as the dominant anion but does not fully neutralise acidity [56]. Total hardness ranges from 13.2 to 174.4 mg/L, with a mean of 71.82 mg/L, classifying the water as predominantly soft to moderately hard. Calcium (0.8–52.58 mg/L; mean 17.32 mg/L) and magnesium (2.04–12.51 mg/L; mean 6.93 mg/L) concentrations support this classification and indicate limited contributions from carbonate or mafic mineral dissolution. Hardness at these levels is not considered a health concern, although it may influence domestic water use through scaling and soap consumption [57]. The dominance of Ca²⁺ and Mg²⁺ over Na⁺ and K⁺ further suggests a hydrochemical facies trending toward Ca–HCO₃ or mixed Ca–Na–HCO₃ types, typical of fresh recharge waters undergoing early-stage geochemical evolution. Major anions, including chloride (2.08–28.29 mg/L; mean 9.40 mg/L) and sulphate (0.50–25.11 mg/L; mean 10.11 mg/L), occur at low concentrations, indicating minimal influence from saline sources, evaporite dissolution, or extensive anthropogenic inputs such as industrial effluents. Similarly, nutrient species show generally low levels. Nitrate–N ranges from 0.01 to 4.25 mg/L (mean 0.89 mg/L), suggesting limited diffuse agricultural or sewage contamination at the scale captured by the summary statistics. These low concentrations may reflect effective dilution by recharge, plant uptake, or denitrification within reducing microenvironments [37, 57]. Nitrite–N and phosphate–P also remain low overall, although their presence indicates that localised anthropogenic inputs cannot be entirely excluded.

Fluoride concentrations (0.00–0.29 mg/L; mean 0.17 mg/L) are uniformly low and well below the WHO guideline value of 1.5 mg/L, implying limited interaction with fluoride-bearing minerals or effective dilution by meteoric recharge. This pattern is consistent with many Ghanaian aquifers that are underlain by F-poor lithologies and contrasts with high-fluoride provinces reported elsewhere in Africa [57]. Trace metal statistics reveal iron and manganese as the most significant water quality constraints. Iron ranges from 0.01 to 5.94 mg/L, with a mean of 2.25 mg/L, while manganese ranges from 0.01 to 1.18 mg/L (mean 0.30 mg/L). These values exceed commonly cited aesthetic and operational guideline levels in a substantial proportion of samples and are indicative of reducing conditions, dissolution of Fe–Mn oxyhydroxides, or mobilisation from lateritic and weathered materials [56, 57]. The co-occurrence of elevated Fe and Mn with low nitrate and sulphate supports a redox-controlled mobilisation mechanism rather than direct anthropogenic metal inputs. Other trace metals, including Cu and Zn, occur at low concentrations and are unlikely to pose significant health risks under current conditions.

4.1.2 Multivariate statistics

Figure 2 presents a Spearman rank correlation heatmap for key physicochemical parameters and composite pollution indices reported for hydrochemical studies in the study

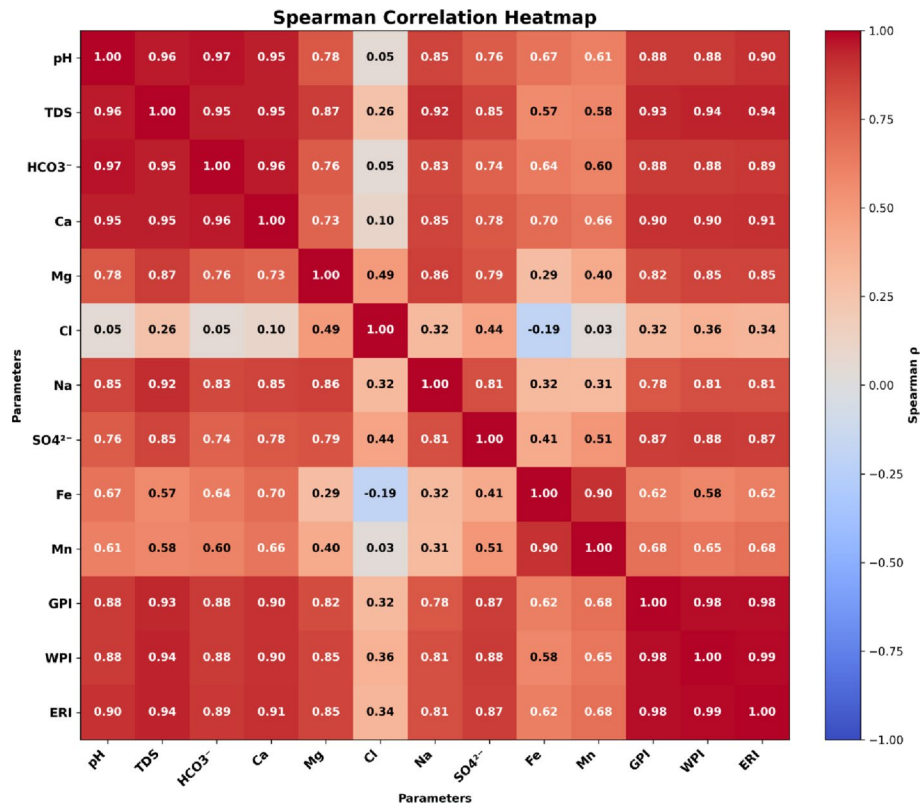


Fig. 2 Correlation matrix of the analysed physicochemical parameters and indices

area. The use of Spearman correlation is appropriate in this context, as hydrochemical datasets in tropical basement terrains are commonly non-normally distributed and influenced by episodic recharge and anthropogenic perturbations [56]. A significant feature of Fig. 2 is the consistently strong positive correlations among pH, TDS, HCO₃⁻, Ca²⁺, Na⁺, Mg²⁺, and SO₄²⁻, with correlation coefficients generally exceeding 0.75. In particular, pH exhibits near-perfect associations with HCO₃⁻ ($r=0.97$), TDS ($r=0.96$), and Ca²⁺ ($r=0.95$). These relationships indicate a hydrochemical regime dominated by water–rock interaction processes, especially carbonate dissolution and silicate weathering, which are well documented as primary controls on groundwater chemistry within Ghana's crystalline basement and Birimian terrains [20, 27]. The tight coupling between TDS and the major cations further confirms that overall mineralisation is driven by ionic enrichment rather than evaporative concentration alone.

The strong association between HCO₃⁻ and alkaline earth metals, particularly Ca²⁺ ($r=0.96$) and Mg²⁺ ($r=0.76$), suggests active dissolution of carbonate phases and weathering of mafic silicates. Similar correlation structures have been reported across Ghanaian aquifers, where bicarbonate-rich Ca–Mg water types dominate recharge and shallow flow systems [59, 60]. The positive correlation between Na⁺ and SO₄²⁻ ($r=0.81$) may additionally reflect contributions from ion exchange processes and localised inputs from anthropogenic sources such as agrochemicals or mine-related effluents in mineralised belts [61]. Chloride shows comparatively weak correlations with most parameters, including pH ($r=0.05$) and HCO₃⁻ ($r=0.05$), and only modest associations with Na⁺ ($r=0.32$) and SO₄²⁻ ($r=0.44$). This pattern suggests that Cl⁻ is not a dominant geogenic control in the datasets and may instead reflect diffuse atmospheric deposition, minor

anthropogenic inputs, or localized contamination rather than systematic water–rock interaction [62]. The weak or inconsistent behaviour of chloride is a recurrent feature in inland Ghanaian aquifers, where marine influence is minimal outside coastal settings. Redox-sensitive elements display a distinct correlation structure. Iron and manganese exhibit a very strong positive relationship ($r=0.90$), consistent with their coupled mobilisation under reducing conditions. Such behaviour is typical of basement aquifers where fluctuating redox conditions control the dissolution of Fe–Mn oxyhydroxides, particularly in zones of low dissolved oxygen or organic matter enrichment [63, 64]. Moderate correlations of Fe and Mn with pH and major ions suggest secondary control by mineral dissolution and sorption–desorption processes rather than purely conservative transport.

The composite indices, including GPI, WPI, and ERI, show near-perfect correlations with one another ($r=0.98$ – 0.99) and strong associations with TDS, pH, Ca^{2+} , and SO_4^{2-} . This reflects their shared dependence on the same underlying hydrochemical variables and confirms that index-based assessments in Ghanaian hydrochemical studies largely respond to overall mineralisation and major ion chemistry. While such coherence supports internal consistency, it also highlights the need for caution in interpretation, as highly correlated indices may offer limited independent information on contamination sources or processes [65].

Principal Component Analysis was employed to assess the primary processes controlling water chemistry and associated pollution and risk signals across the area. Three principal components with eigenvalues greater than unity were retained, together explaining 94.3% of the total variance (Table 2). This high cumulative variance indicates that the extracted components adequately represent the dominant hydrochemical processes influencing groundwater composition within the study area. Such a strong variance explanation is commonly observed in hydrochemical datasets where a limited number of geochemical processes exert primary control over groundwater evolution.

PC1 explains 71.2% of the variance and is characterised by very strong positive loadings for HCO_3^- , TDS, Ca^{2+} , conductivity and pH, with additional strong contributions from Na^+ . Moderate positive loadings were also observed for SO_4^{2-} and Mg^{2+} . This

Table 2 Rotated component matrix showing principal factor loadings of hydrochemical parameters and pollution indices

Hydrochemical parameter	PC1	PC2	PC3
HCO_3^-	0.988	0.058	0.107
TDS	0.964	0.217	0.124
Ca^{2+}	0.957	0.110	0.083
Conductivity	0.951	0.216	0.180
pH	0.937	0.095	−0.026
Na^+	0.833	0.351	0.369
SO_4^{2-}	0.704	0.622	−0.034
Mg^{2+}	0.670	0.607	0.208
Cl^-	0.022	0.977	0.017
K^+	0.120	0.030	0.987
<i>PCA statistics</i>			
Eigenvalue	7.118	1.281	1.032
Variance explained (%)	71.181	12.806	10.323
Cumulative variance (%)	71.181	83.988	94.311

Extraction method: Principal Component Analysis (PCA). Rotation method: Varimax with Kaiser normalization

Rotation converged in 4 iterations. Bold values indicate strong factor loadings (>0.70)

component represents the principal groundwater mineralisation and water-rock interaction process controlling hydrochemical evolution within the aquifer system [64]. The strong association among bicarbonate, calcium, conductivity and dissolved solids reflects carbonate dissolution, silicate weathering and progressive ion enrichment during groundwater circulation through the regolith and fractured basement aquifers. The loading of pH further indicates that mineral dissolution and buffering reactions exert important controls on groundwater chemistry [32].

PC2 accounts for 12.8% of the total variance and is dominated by a very strong positive loading for Cl^- , together with moderate positive loadings for SO_4^{2-} and Mg^{2+} . This component reflects secondary salinity enrichment and localised hydrochemical variability associated with chloride and sulphate contributions. The strong chloride loading may indicate local anthropogenic influence, evaporative concentration of domestic wastewater inputs, agricultural return flow or dissolution of chloride-bearing minerals. The association of sulphate within this component further supports the contribution of sulphate-bearing minerals and possible sulphide oxidation processes identified in earlier hydrochemical interpretations [65, 66].

PC3 explains 10.3% of the total variance and is strongly associated with K^+ . This component likely reflects localised potassium enrichment arising from silicate mineral weathering, ion-exchange reactions, or agricultural inputs. The isolation of potassium within a separate component suggests that its hydrochemical behaviour is partly independent of the dominant groundwater mineralisation processes represented by PC1 and PC2.

Overall, the PCA results confirm that groundwater chemistry within the study area is predominantly controlled by water-rock interaction processes, with secondary influences from localised salinity enrichment and anthropogenic inputs.

To further evaluate the spatial relationships among hydrochemical variables and groundwater samples, a PCA biplot was constructed using the first two principal components (Fig. 3). The biplot provides a multivariate visualisation of parameter loadings and community-level groundwater variability across Juaso, Segyemase and Nsuapemso. The first two principal components account for 84.0% of the total variance, with PC1 and PC2 explaining 71.2% and 12.8%, respectively, indicating that the dominant hydrochemical processes are adequately represented within the ordination space [62].

PC1 represents the principal hydrochemical evolution axis and is strongly associated with TDS, Ca^{2+} , HCO_3^- , pH and Na^+ . The close grouping of these variables reflects progressive groundwater mineralisation through carbonate dissolution, silicate weathering and prolonged water-rock interaction. The strong association between EC and TDS indicates increasing ionic strength and dissolved solids accumulation, while the clustering of Ca^{2+} and HCO_3^- further supports the influence of carbonate dissolution and prolonged interaction between groundwater and aquifer mineral weathering [64]. These relationships corroborate the hydrochemical facies identified in the Piper diagram and the rock-weathering dominance observed in the Gibbs plots. Although bicarbonates dominate facies prevail in several groundwater samples, increasing sulphate and chloride contributions in some locations indicate progressive hydrochemical evolution toward mixed Ca-Mg- SO_4 facies, particularly in areas influenced by mining activities and prolonged water-rock interaction.

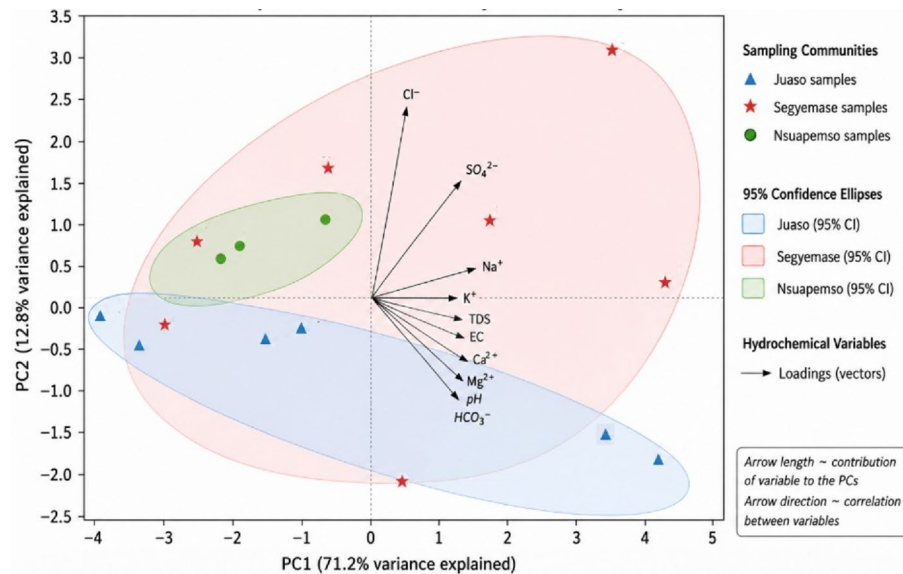


Fig. 3 PCA biplot showing the distribution of groundwater samples and hydrochemical parameters loadings across the Juaso, Segyemase and Nsuapemso communities. PC1 and PC2 collectively explain 84.0% of the total variance

PC2 is primarily influenced by Cl^- and SO_4^{2-} , suggesting secondary hydrochemical controls related to sulphate mineral, sulphide oxidation and localised chloride enrichment. The relatively strong Cl^- loading indicates that chloride contributes significantly to hydrochemical variability across the study area and may reflect local anthropogenic influence, evaporative concentration, or dissolution of chloride-bearing minerals [67]. Similarly, the positive association of SO_4^{2-} with PC2 supports earlier interpretations from the Piper and saturation index analyses, which indicated the contribution of sulphate-bearing minerals and sulphide oxidation processes.

Community-wise clustering within the biplot reveals measurable spatial variability in groundwater chemistry among the sampled communities. Juaso samples generally plot along moderate-to-high PC1 values and negative PC2 scores, reflecting relatively evolved groundwater characterised by higher mineralisation and stronger influence of carbonate weathering and dissolved solids [66]. The relatively broad Juaso confidence ellipse indicates greater hydrochemical heterogeneity within the community. Segyemase samples display the widest dispersion across the ordination space, suggesting mixed hydrochemical signatures, variable groundwater residence times, and multiple geochemical controls. Some Segyemase samples show a stronger association with the Cl^- and SO_4^{2-} vectors, indicating localised sulphate and chloride enrichment [68]. In contrast, Nsuapemso samples form a comparatively tighter cluster, indicating relatively homogeneous groundwater chemistry and less variable hydrochemical evolution pathways.

The PCA biplot provides an integrated multivariate confirmation of the hydrochemical processes identified from the Piper, Gibbs, ion-exchange and saturation index analyses, demonstrating that groundwater chemistry within the study area is predominantly controlled by water-rock interaction, carbonate dissolution, silicate weathering, groundwater mineralisation and localised sulphate-chloride enrichment processes [65].

4.1.3 Mechanism controlling groundwater geochemistry in the area

The Piper diagram shows a clear and consistent hydrochemical pattern across the sampled waters (Fig. 4). In the cation triangle, most samples plot toward the Ca^{2+} corner, with a noticeable but secondary contribution from Mg^{2+} , while $\text{Na}^+ + \text{K}^+$ remain comparatively low. This distribution indicates a dominance of alkaline earth metals over alkali metals, suggesting that groundwater chemistry is primarily controlled by rock–water interaction rather than by ion exchange or saline inputs. In the anion triangle, most samples plot toward the $\text{SO}_4^{2-} - \text{Cl}^-$ side, with very limited influence from the $\text{HCO}_3^- + \text{CO}_3^{2-}$ apex, indicating dominance of strong-acid anions over carbonate alkalinity. Several samples display a distinct chloride contribution, suggesting mixed sulphate-chloride waters rather than purely sulphate-type groundwater. The subdued bicarbonate signature suggests a limited influence of recent recharge, dominated by carbonic acid weathering. In the central diamond, the samples cluster mainly within the $\text{Ca-Mg-SO}_4\text{-Cl}$ field, defining a calcium-magnesium-sulphate-chloride hydrochemical facies consistent with evaporite dissolution and water-rock interaction processes. This facies reflects chemically evolved groundwater and is characteristic of systems influenced by the dissolution of sulphate-bearing minerals such as gypsum or anhydrite, sulphide oxidation processes and additional chloride inputs potentially derived from halite dissolution, evaporative concentration or anthropogenic activities. The samples from the three communities exhibit broadly similar clustering within the $\text{Ca-Mg-SO}_4\text{-Cl}$ facies, indicating that groundwater chemistry across the study area is governed by consistent regional-scale process. However, subtle variations are evident, with Segyemase samples showing slightly greater dispersion within the diamond field, suggesting localised influences possibly related to mining activities or land-use practices. In contrast, Juaso and Nsuapemso samples display relatively tighter clustering, indicating more uniform hydrochemical

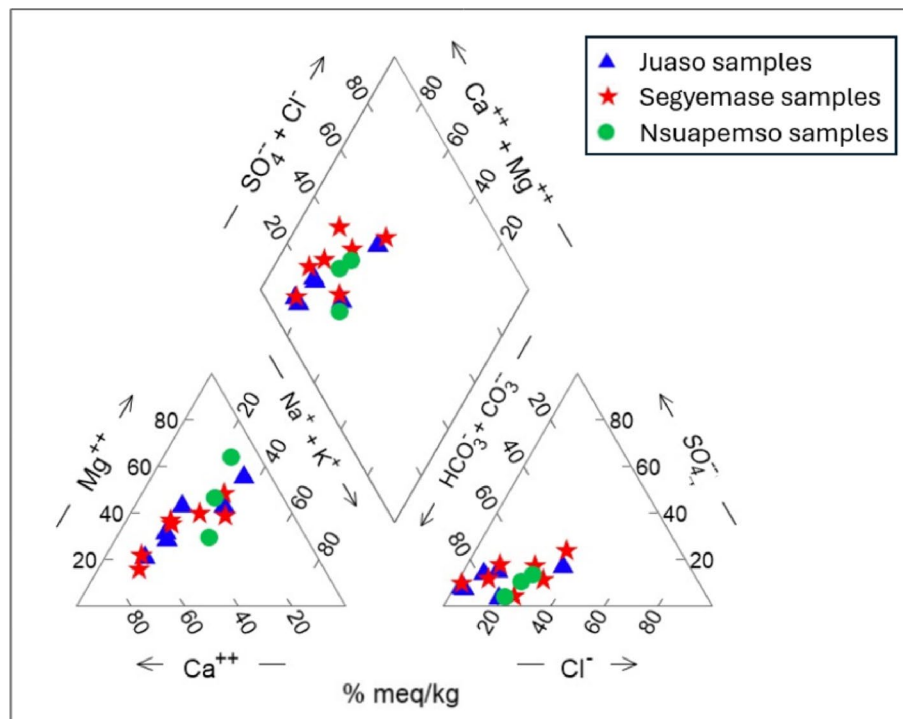


Fig. 4 Piper plot of groundwater in the study area

conditions. These observations highlight that while the dominant geochemical processes are consistent across the study area, local-scale factors contribute to minor spatial variability in groundwater composition. Such processes are commonly associated with mineralized terrains and areas affected by mining activities, reinforcing the influence of both regional geology and localised anthropogenic inputs on groundwater evolution in the study area [1, 69].

The Gibbs diagrams (Fig. 5a & b) were used to provide insight into the dominant processes controlling groundwater chemistry in the study area. In both plots, most samples fall within the rock weathering dominance field, with total dissolved solids generally ranging from moderate to elevated levels. This distribution indicates that groundwater composition is primarily governed by water–rock interaction rather than by atmospheric precipitation or strong evaporative concentration. In the $\text{Na}^+ / (\text{Na}^+ + \text{Ca}^{2+})$ plot, samples cluster at intermediate ratios, suggesting limited alkali enrichment and reinforcing the dominance of calcium-bearing minerals. This pattern points to silicate and, to a lesser extent, carbonate weathering as key contributors to dissolved ions. The absence of clustering toward higher ratios implies that cation exchange and saline inputs play only a minor role in controlling groundwater chemistry.

Similarly, in the $\text{Cl}^- / (\text{Cl}^- + \text{HCO}_3^-)$ plot, most samples plot at low to moderate chloride fractions while remaining within the rock weathering domain. This reflects a hydrochemical system where bicarbonate and sulphate sources linked to mineral dissolution outweigh atmospheric or marine-derived chloride contributions. A few samples show slight shifts toward higher TDS values, indicating localised evaporation effects, but these do not dominate the overall system.

The samples from the three communities consistently plot within the rock weathering dominance field, indicating that the primary control on groundwater chemistry is uniform across, showing marginally higher TDS values and a broader spread, suggesting localised solute enrichment possibly linked to mining or land-use activities [70]. In contrast, Juaso and Nsuaipemso samples exhibit relatively tighter clustering at lower to moderate TDS values, reflecting more stable hydrochemical conditions. These patterns indicate that while regional-scale water–rock interaction dominates groundwater evolution, local factors contribute to minor spatial variability [56, 71].

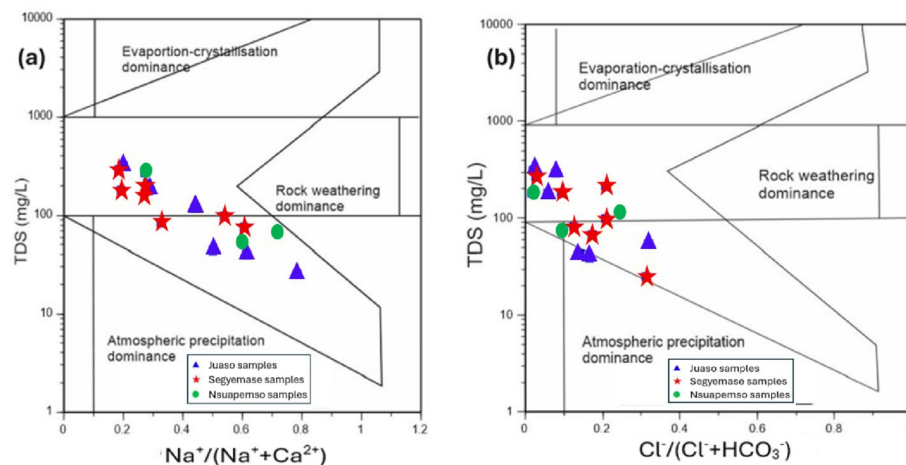


Fig. 5 Gibbs [70] plots, (a) TDS (mg/L) vs. $\text{Na}^+ / (\text{Na}^+ + \text{Ca}^{2+})$ (b) TDS (mg/L) vs. $\text{Cl}^- / (\text{Cl}^- + \text{HCO}_3^-)$

The Gibbs diagrams, therefore, confirm that groundwater chemistry in the study area is primarily controlled by water-rock interactions, particularly silicate and subordinate carbonate weathering [27, 68]. The observed low to moderate $\text{Na}^+(\text{Na}^+ + \text{Ca}^{2+})$ and $\text{Cl}^-(\text{Cl}^- + \text{HCO}_3^-)$ ratios, together with moderate TDS values, indicate limited evaporative concentration and minimal influence of atmospheric or saline inputs. These findings are consistent with the hydrochemical facies identified in the Piper diagram and highlight the dominance of geogenic processes with minor localised anthropogenic influence [72].

The ion-ion plots (Fig. 6) collectively illustrate the role of cation exchange and reverse ion exchange in regulating groundwater chemistry [71, 73]. The 1:1 equiline in each diagram represents chemical equivalence between the paired ions, while deviations from this line reflect exchange reactions between groundwater and aquifer materials [70]. In the Na^+ versus Cl^- plot, most samples plot below the equiline, indicating an excess of chloride relative to sodium. This pattern suggests that sodium is being removed from solution through ion exchange processes, where Na^+ in groundwater is adsorbed onto exchange sites in the aquifer matrix and replaced by divalent cations, particularly Ca^{2+} and Mg^{2+} [73]. This behaviour is typical of normal cation exchange in clay-rich or weathered aquifer systems. The Ca^{2+} versus Mg^{2+} relationship further supports this interpretation. Several samples deviate from the equiline, showing variable enrichment of calcium relative to magnesium [66]. This may reflect differential release of Ca^{2+} during gypsum/anhydrite dissolution and ion exchange reactions, while magnesium may be partly retained on exchange sites or influenced by slower dolomite dissolution kinetics [67, 72, 74].

A strong positive relationship between $(\text{Ca}^{2+} + \text{Mg}^{2+})$ and $(\text{HCO}_3^- + \text{CO}_3^{2-})$ is observed, with many samples plotting near the equiline. This suggests that carbonate mineral dissolution contributes to alkaline earth cations in the groundwater [73]. However, considering the dominance of sulphate facies identified in the Piper diagram, carbonate weathering appears to play a secondary role relative to sulphate-bearing mineral

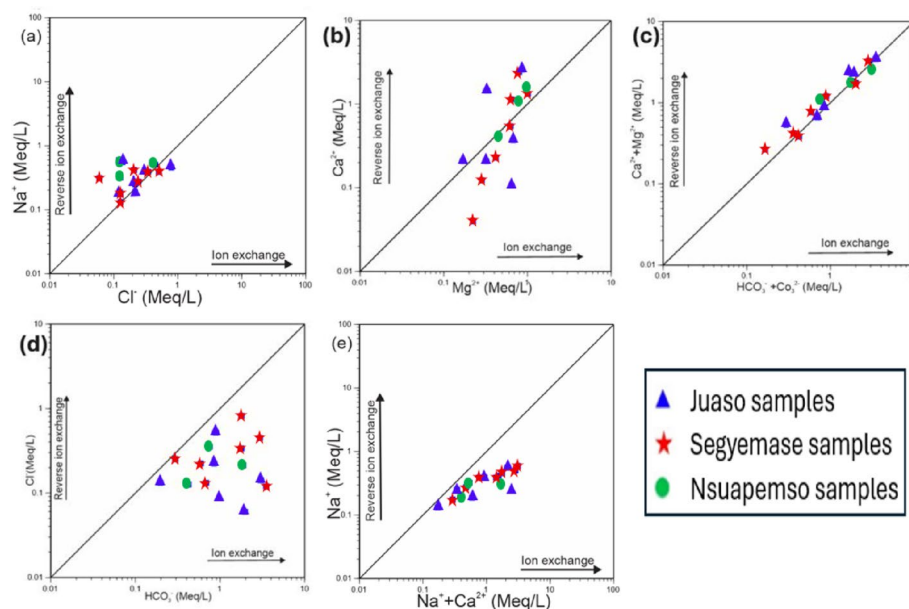


Fig. 6 Major ion relationships showing processes that control groundwater chemistry. **(a)** Na^+ (Meq/L) vs. Cl^- (Meq/L) **(b)** Ca^{2+} (Meq/L) vs. Mg^{2+} (Meq/L) **(c)** $\text{Ca}^{2+} + \text{Mg}^{2+}$ (Meq/L) vs. $\text{HCO}_3^- + \text{CO}_3^{2-}$ (Meq/L) **(d)** Cl^- (Meq/L) vs. HCO_3^- (Meq/L) **(e)** Na^+ (Meq/L) vs. $\text{Na}^+ + \text{Ca}^{2+}$ (Meq/L)

dissolution [68, 73]. The proximity to the equiline also suggests limited disturbance of this balance by extensive ion exchange in some samples [70]. The Cl^- versus HCO_3^- plot shows most data points displaced toward higher bicarbonate concentrations relative to chloride. This pattern reflects a transition from recharge-dominated, bicarbonate-rich waters toward more chemically evolved compositions [67]. It also further corroborates that chloride behaves conservatively, while bicarbonate is actively generated through silicate and carbonate weathering processes [74, 75].

The Na^+ versus $(\text{Na}^+ + \text{Ca}^{2+})$ relationship provides clear evidence of reverse ion exchange [68]. Most samples plot well above the equiline, indicating enrichment of sodium relative to calcium [72]. This suggests that Ca^{2+} in groundwater is being adsorbed onto exchange sites, while Na^+ is released into solution from the aquifer matrix [73]. Such reverse ion exchange is commonly associated with prolonged water-rock interaction, clay mineral surfaces, and evolving groundwater residence times [70, 74]. The ion-ion relationships show broadly consistent trends across Segyemase, Juaso and Nsuapemso, indication that the underlying geochemical processes are similar throughout the study area. However, Segyemase samples display slightly greater dispersion across several plots, suggesting localised variability possibly linked to anthropogenic activities such as mining or land use. In contrast, Juaso and Nsuapemso samples exhibit relatively tighter clustering, reflecting more stable hydrochemical conditions. These observations indicate that while ion exchange and reverse ion exchange processes operate across all communities, their intensity and expression vary slightly at the local scale. Overall, the ion-ion plots confirm that groundwater evolution in the study area is governed by a combination of cation exchange, reverse ion exchange and mineral dissolution processes.

Mineral saturation indices computed using PHREEQC indicate that all examined mineral phases (calcite, dolomite, gypsum, halite and fluorite) have negative SI values, demonstrating that the groundwater is undersaturated with respect to these minerals. This geochemical condition implies that precipitation of these minerals is not occurring and that water-rock interaction is dominated by ongoing dissolution under open-system conditions. Calcite shows a strong positive relationship between SI and the combined concentration of Ca^{2+} and HCO_3^- ($r \approx 0.88$), demonstrating that carbonate weathering is a primary mechanism regulating calcium and alkalinity in the system (Fig. 7a). As Ca^{2+} and bicarbonate increase, groundwater approaches equilibrium with calcite, reflecting progressive interaction with carbonate-bearing lithologies or secondary carbonate phases. However, the persistence of negative SI values indicates that full saturation is not attained, likely due to continuous recharge and dilution. A similar but slightly subdued trend is observed for dolomite, where SI correlates strongly with $\text{Ca}^{2+} + \text{Mg}^{2+} + \text{HCO}_3^-$ ($r \approx 0.85$) (Fig. 7b). This pattern suggests that dolomite dissolution contributes to the supply of both calcium and magnesium, although equilibrium is approached more slowly than for calcite. This behaviour is consistent with the known kinetic constraints on dolomite dissolution relative to calcite and supports a carbonate-dominated hydro-geochemical framework.

The gypsum saturation index exhibits a clear positive association with $\text{Ca}^{2+} + \text{SO}_4^{2-}$ ($r \approx 0.82$) (Fig. 7c), indicating that sulphate concentrations are influenced by gypsum dissolution or sulphate release from related sources such as sulphide oxidation. Despite this relationship, the groundwater remains markedly undersaturated with respect to gypsum, suggesting limited evaporite abundance or effective dilution by recharge. Halite displays

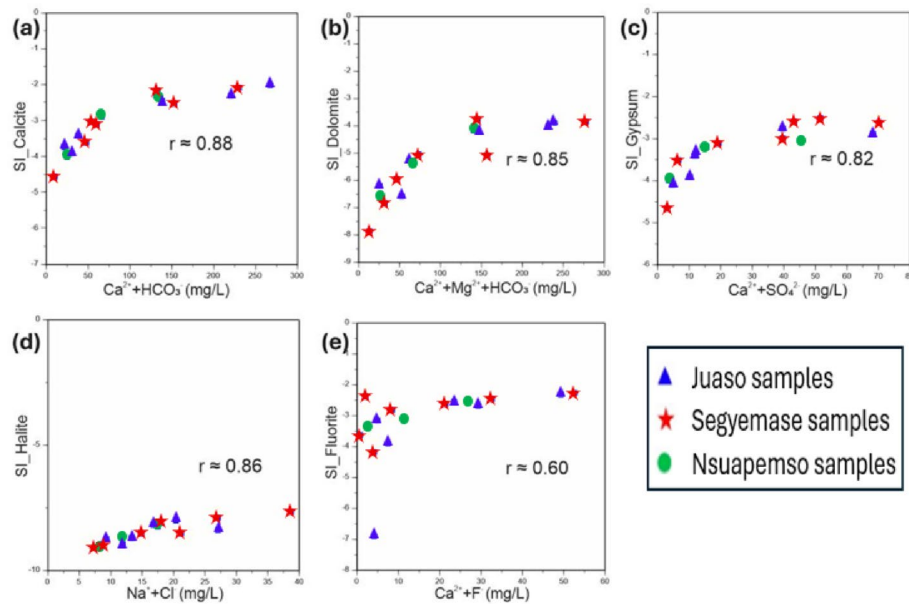


Fig. 7 Plots showing the relationship between mineral phases and ions it in the groundwater (a) SI_Calcite vs. $\text{Ca}^{2+}(\text{mg/L}) + \text{HCO}_3^{-}(\text{mg/L})$ (b) SI_Dolomite vs. $\text{Ca}^{2+}(\text{mg/L}) + \text{Mg}^{2+}(\text{mg/L}) + \text{HCO}_3^{-}(\text{mg/L})$ (c) SI_Gypsum vs. $\text{Ca}^{2+}(\text{mg/L}) + \text{SO}_4^{2-}(\text{mg/L})$ (d) SI_Halite vs. $\text{Na}^{+}(\text{mg/L}) + \text{Cl}^{-}(\text{mg/L})$ (e) SI_Fluorite vs. $\text{Ca}^{2+}(\text{mg/L}) + \text{F}^{-}(\text{mg/L})$

a strong linear relationship between SI and $\text{Na}^{+} + \text{Cl}^{-}$ ($r \approx 0.86$) (Fig. 7d), reflecting the role of halite dissolution and/or evaporative concentration in controlling salinity. The consistently negative SI values indicate that halite precipitation is absent, and sodium–chloride enrichment is driven by dissolution processes rather than closed-basin evaporation. In contrast, fluorite shows only a moderate correlation between SI and $\text{Ca}^{2+} + \text{F}^{-}$ ($r \approx 0.60$) (Fig. 7e), implying that fluoride behaviour is governed by more complex and multi-process controls. The weak approach to fluorite equilibrium suggests that fluoride concentrations are influenced not only by fluorite dissolution but also by competing processes such as silicate weathering, ion exchange, and variations in calcium availability, which can suppress fluorite saturation. The relationships between saturation indices and controlling ions show consistent trends across Segyemase, Juaso and Nsuapemso, indicating that mineral dissolution processes are regionally uniform [62]. However, Segyemase samples display slightly greater variability in SI values and ion concentrations, suggesting localised influences possibly related to mining activities or land-use practices [13]. In contrast, Juaso and Nsuapemso samples exhibit relatively tighter clustering, reflecting more stable hydrogeochemical conditions. These patterns indicate that while dissolution processes dominate across the study area, local factors contribute to minor spatial variability in groundwater evolution. The consistently negative saturation indices confirm that groundwater chemistry is controlled by active mineral dissolution under open-system conditions, with no evidence of mineral precipitation [32].

4.1.4 Suitability for drinking water

To ensure public health safety, it is critical to assess the suitability of groundwater for drinking water. This study employed the Water Quality Index (WQI) to evaluate the chemical quality of groundwater from the sampled boreholes [52]. The WQI integrates multiple physicochemical parameters, including TDS, pH, total hardness, major cations

(Ca^{2+} , Mg^{2+} , Na^+ , K^+), and anions (Cl^- , HCO_3^- , SO_4^{2-} , NO_3^- , F^-) into one index that gives a reflection of the water quality. The calculated WQI of the samples was classified using the scheme detailed in Sect. 3.3.4, with values categorized as excellent (< 50) and unsuitable (> 300). All the samples were in the “excellent” category (Table 3), indicating that the groundwater, from a chemical standpoint, is highly suitable for drinking. The low WQI values reflect the favourable physicochemical characteristics of the samples were within the limits of the WHO guidelines [41]. The WQI-based assessment demonstrated that the groundwater in the studied communities is safe for domestic consumption chemically, confirming the suitability of these sources for drinking, provided that microbiological safety is ensured.

4.2 Water quality assessment

The suite of indexical proxies applied here captures different dimensions of groundwater quality (Table 4), so convergence across indices strengthens interpretation, while divergence highlights process complexity and index sensitivity to parameter selection and weighting [50, 76]. Across the 16 sampled boreholes, the indices show the following central tendencies and ranges: GPI 0.116 to 0.543 (mean 0.303), WPI 0.234 to 1.113 (mean 0.630), CPI 0.018 to 0.133 (mean 0.077), WNPI 0.011 to 1.203 (mean 0.264), TMPI 0.92 to 17.31 (mean 5.96), and ERI 0.102 to 0.512 (mean 0.278). These distributions indicate low to moderate general physicochemical stress by some proxies, but clear trace metal loading by TMPI, with nutrient impairment present at specific locations. This pattern is consistent with Ghanaian groundwater settings where recharge-driven dilution can coexist with localised anthropogenic inputs and geogenic metal mobilisation, especially in shallow, weathered and fractured aquifers that are hydraulically connected to the surface environment [41, 77, 78].

Although some indices exhibited positive correlations due to shared input parameters, each index captures a distinct aspect of groundwater quality. While GPI, CPI and WNPI provide general pollution status, TMPI specifically highlights trace metal enrichment, and ERI assesses potential ecological risk levels. Therefore, the combined interpretation

Table 3 Calculated WQI values for all boreholes

Sample ID	WQI	Classification
JSB 01	0.19516	Excellent
JSB 02	0.532706	Excellent
JSB 03	1.141037	Excellent
JSB 04	1.290936	Excellent
JSB 05	0.164658	Excellent
JSB 06	0.467524	Excellent
SGB 01	0.203758	Excellent
SGB 02	0.186756	Excellent
SGB 03	1.143675	Excellent
SGB 04	0.916018	Excellent
SGB 05	1.317492	Excellent
SGB 06	0.836398	Excellent
SGB 07	0.22456	Excellent
NSB 02	0.208296	Excellent
NSB 03	0.189486	Excellent
NSB 04	0.17519	Excellent

< 50 excellent water, 50–100 good water, 100–200 poor water, 200–300 very poor water, > 300 Water unsuitable for drinking

Table 4 Summary of the calculated indices for evaluating groundwater quality across the area

Sample	GPI	WPI	CPI	WNPI	TMPI	ERI
JSB01	0.253	0.452	0.097	0.051	4.16	0.196
JSB02	0.319	0.671	0.102	0.312	5.14	0.299
JSB03	0.466	0.933	0.115	1.092	15.55	0.433
JSB04	0.295	0.622	0.087	1.203	17.31	0.274
JSB05	0.118	0.234	0.053	0.011	0.92	0.102
JSB06	0.383	0.813	0.133	0.268	8.11	0.368
SGB01	0.148	0.314	0.044	0.024	1.87	0.136
SGB02	0.116	0.243	0.051	0.035	2.03	0.104
SGB03	0.395	0.813	0.073	0.221	6.35	0.364
SGB04	0.502	0.946	0.071	0.233	6.66	0.473
SGB05	0.543	1.113	0.111	0.287	7.94	0.512
SGB06	0.424	0.912	0.063	0.305	7.44	0.39
SGB07	0.274	0.636	0.086	0.065	3.43	0.246
NSB02	0.265	0.576	0.092	0.034	2.95	0.234
NSB03	0.214	0.484	0.043	0.042	3.54	0.194
NSB04	0.136	0.312	0.018	0.038	2.03	0.116

enhances robustness and reduces the likelihood of underestimating localized metal related risks.

4.2.1 Groundwater Pollution Index

Using the stated GPI classes, 4 of 16 samples (25%) fall in low pollution (≤ 0.20), 8 (50%) fall in moderate pollution (0.21 to 0.40), and 4 (25%) fall in high pollution (0.41 to 0.60). No sample reaches the very high class (> 0.60). The high GPI group comprises JSB03, SGB04, SGB05, and SGB06, suggesting spatially clustered deterioration rather than catchment-wide failure. Since GPI aggregates weighted departures across multiple parameters, these outcomes typically reflect combined effects of solute enrichment from water-rock interaction and superimposed anthropogenic additions such as sanitation leakage, agrochemical return flows, and small-scale industrial discharges [45, 77]. In a Ghanaian context, this is plausible in settings with thin regolith cover and shallow water tables, where episodic recharge can rapidly transfer surface-derived solutes into the groundwater system [77, 79].

4.2.2 Water Pollution Index

The WPI classification indicates a stronger impairment signal than GPI. Only 2 samples (12.5%) are clean (≤ 0.30), 4 (25%) are slightly polluted (0.31 to 0.50), 4 (25%) are moderately polluted (0.51 to 0.75), and 6 (37.5%) are heavily polluted (> 0.75). The heavily polluted group includes JSB03, JSB06, SGB03, SGB04, SGB05, and SGB06. This shift toward a poorer status is consistent with the design intent of WPI as an integrated ranking tool that can be more discriminating where multiple parameters depart moderately from guideline levels, rather than requiring extreme exceedances in a few parameters [76, 80]. Therefore, the WPI result supports a narrative of multi-parameter degradation in parts of the aquifer system, even when individual parameters may not all breach drinking water limits. From a hydrogeochemical standpoint, this often reflects mixed control, with baseline mineralisation from silicate weathering and ion exchange providing the chemical envelope, while human activities elevate select solutes within vulnerable recharge and abstraction zones [45, 76, 81].

4.2.3 Water Nutrient Pollution Index

The WNPI outcomes show 8 samples (50%) as unpolluted (≤ 0.10), 0 as slight nutrient pollution (0.11 to 0.20), 6 (37.5%) as moderate nutrient pollution (0.21 to 0.50), and 2 (12.5%) as severe nutrient pollution (> 0.50), specifically JSB03 and JSB04. Nutrient-related impairment in groundwater is most linked to nitrate loading from on-site sanitation, animal wastes, and fertiliser use, with spatial variability controlled by land use intensity, vadose zone thickness, and redox conditions that regulate denitrification efficiency [47, 80, 82]. In Ghana, evidence from basin-scale and community studies also shows that nitrate issues are often localised, reflecting point source sanitation leakage and peri-urban agriculture rather than uniform regional contamination [41, 58, 79]. The WNPI pattern here aligns with that expectation, where most locations remain low, but specific boreholes show clear nutrient stress that warrants targeted source tracing and protection zoning.

4.2.4 Comprehensive Pollution Index

All CPI values fall in the clean category (≤ 0.20), with a narrow range of 0.018 to 0.133. This is a key interpretive hinge because CPI is often used to summarise basin-scale conditions, but it is sensitive to the choice of parameters included and the way aggregation dampens extremes [21]. The apparent contradiction between universally clean CPI and the degraded WPI profile indicates that CPI, as implemented here, either prioritises a subset of parameters that remain low, applies a scaling that compresses variability, or averages across constituents in a way that masks multi-parameter moderate deviations.

4.2.5 Trace Metals Pollution Index

TMPI shows the most unambiguous signal. 14 of 16 samples (87.5%) classify as very high metal pollution (> 2.00). The remaining two comprise JSB05 (high, 0.92 to 2.00 under your scheme, which falls below 1.01 to 2.00, so it sits at the high end of “moderate” if strictly applied; check threshold application) and SGB01 (high, 1.87). Regardless of this boundary detail, the overall pattern indicates pervasive trace metal enrichment relative to the adopted reference frame. In Ghana, such enrichment can reflect geogenic contributions from mineralised lithologies and weathering of sulphide-bearing assemblages, as well as anthropogenic sources including mining-related releases, waste disposal, and small-scale industrial activity [41, 77, 80].

4.2.6 Ecological Risk Index

All ERI values fall far below the lowest risk threshold (< 40), placing 100% of samples in the low ecological risk class. This result contrasts sharply with the TMPI outcome. Two interpretations are plausible. First, ERI as computed here may use toxicity weighting, background values, or scaling that substantially down weights the metals driving TMPI, producing low composite risk despite elevated enrichment. Second, the groundwater metal signal may involve elements or species with lower ecological toxic response within the ERI formulation, or low assumed bioavailability. Enrichment tells you, “how much above baseline.” Ecological risk tells you “how harmful under assumed toxicity and exposure.” The divergence, therefore, highlights methodological sensitivity and supports the argument for reporting both, alongside clear disclosure of ERI parameterisation and its suitability for groundwater contexts.

Table 5 Descriptive statistics of microbial parameters

Parameter	Minimum	Maximum	Mean	Median	Std. Dev
TC	5	2790	554	47	823
FC	0	1302	106	0	252
<i>E. Coli</i>	0	1116	87	0	221
<i>Salmonella</i>	0	0	0	0	0
<i>Vibrio</i>	0	0	0	0	0

Table 6 Percentage exceedance of WHO Guidelines

Parameter	Sample above guidelines	Total samples	%Exceedance
TC	13	16	81.3
FC	2	16	12.5
<i>E. coli</i>	2	16	12.5
<i>Salmonella</i>	0	16	0
<i>Vibrio</i>	0	16	0

4.3 Microbiological quality of groundwater

Microbial contamination was evaluated using standard indicators and pathogenic organisms. Total coliform (TC), Faecal coliform (FC), *Escherichia coli* (*E. coli*), *Salmonella spp.*, and *Vibrio spp.* were assessed. The evaluation involved descriptive statistics, guideline exceedance analysis, microbial risk classification and quantitative microbial risk assessment to provide a meaningful basis for evaluating microbial safety and its implications on human health.

4.3.1 Descriptive statistics and microbial counts

Microbial concentrations varied widely across boreholes; TC ranged between 5 and 2,790 CFU/100 mL with an average of 554 CFU/100 mL (Table 5). FC and *E. coli* were detected in two boreholes, with minimum counts of 36 and 26 CFU/100 mL, maximum counts of 1202 and 1116 CFU/100 mL, respectively. *Salmonella spp.* and *Vibrio spp.* were, however, not detected, which indicated limited immediate pathogen risk [54, 83].

Variation in microbial quality reflects respective local borehole conditions, proximity to contamination sources and general surrounding sanitation. Boreholes with higher total suspended solids, turbidity, or near a potential source of pollution had a corresponding higher microbial count, consistent with similar studies [21, 84, 85]. Previous studies in Ghana and Burkina Faso reported similar patterns, emphasizing a relationship between microbial water quality and environmental factors [15, 16, 83].

4.3.2 Percentage exceedance of WHO Guidelines

Most of the boreholes, except three, exceeded the 0 CFU/100 mL limit recommended by WHO. The exceedance analysis reveals that 81% of boreholes exceeded WHO guidelines [41] for TC, and 12.5% of the boreholes had samples with FC and *E. coli* limits above those recommended by WHO (Table 6). The widespread TC contamination generally shows microbial pollution, while the low observation of FC and *E. coli* indicates localised high-risk sites (JSB 05 and JSB 06).

4.3.3 Risk classification and Microbial Contamination Index (MCI)

Based on TC and *E. coli* counts, the boreholes were categorized into low, medium, and high-risk levels. With MCI values of 1488 (TC), 1302 (FC), 1116 (*E. coli*), and 1116 (TC),

Table 7 Risk classification and MCI

Sample	TC risk	E. coli risk	MCI (TC/FC/E. coli.)
JSB 05	High	High	1488/1302/1116
JSB 06	High	High	1116/36/26
Others	Low-high	Low	0–2790/0/0

Table 8 QMRA for *E. coli*

Sample	E. coli (CFU/100 mL)	Daily Dose (CFU/day)	Infection Probability P_{inf}	Annual Probabil- ity P_{annual}
JSB 05	1116	22,320	0.00235	0.58 (58%)
JSB 06	26	520	0.000055	0.02 (2%)

36 (FC), 26 (*E. coli*), JSB 05 and JSB 06 were classified as high-risk (Table 7). Other boreholes, despite high counts of showed low to undetected FC and *E. coli*, suggesting recent faecal contamination was limited. The MCI identifies boreholes that require immediate attention and informs prioritisation for remediation and monitoring [54, 83].

4.3.4 Quantitative Microbial Risk Assessment (QMRA)

Daily doses and infection probabilities of *E. coli* were calculated with the Beta-Poisson model [55]. JSB 05 had a daily dose of 22,320 CFU/day, giving an annual infection probability of 58%, while JSB 06 had a daily dose of 520 CFU/day with an annual infection probability of 2% (Table 8). High-risk boreholes, specifically JSB 05, pose significant public health risks. Even low levels of contamination (JSB 06) can be hazardous in the long term if consumed frequently.

4.4 Integrated drinking water suitability assessment

4.4.1 Chemical quality and hydrogeochemistry

Generally, the groundwater indicated low ionic concentrations (all major cations and anions), trace elements and physical parameters were within the WHO guidelines limits for drinking water. The WQI values computed for all 16 boreholes ranged from 0.165 to 1.318, classifying all sources as excellent water (< 50). The computed pollution indices were also well within the safe limits, indicating that the water is not chemically polluted. These findings align with previous studies in rural and peri-urban Ghana, where good physicochemical parameters and low trace element concentrations were reported in groundwater and attributed to limited anthropogenic impact and favourable hydrogeochemical conditions [86, 87].

4.4.2 Microbiological quality and public health implications

Despite excellent chemical quality, microbiological assessment revealed a widespread total coliform (TC) contamination, with counts as high as 2790 CFU/100 mL. Faecal coliform and *E. coli* were detected in JSB 05 and JSB 06, indicating recent faecal contamination and high risk for waterborne infections. *Salmonella spp.* and *Vibrio spp.*, were not detected in any of the samples, suggesting limited immediate risk from these pathogens [88]. Microbial contamination correlated with physicochemical parameters and local sanitary conditions, such as turbidity, total suspended solids, and proximity to latrines or animal waste. These results confirm that microbial hazards in groundwater

Table 9 Integrated drinking water suitability and prioritization of boreholes for remediation

Sample ID	WQI	Classification	Microbial risk	Suitability	Priority for remediation
JSB 01	0.20	Excellent	Low	Safe	Low
JSB 02	0.53	Excellent	Low	Safe	Moderate
JSB 03	1.14	Excellent	Low	Safe	Low
JSB 04	1.29	Excellent	Low	Safe	Moderate
JSB 05	0.16	Excellent	High	Unsafe	High
JSB 06	0.47	Excellent	High	Unsafe	High
SGB 01	0.20	Excellent	Low	Safe	Moderate
SGB 02	0.19	Excellent	Low	Safe	Moderate
SGB 03	1.14	Excellent	Low	Safe	Moderate
SGB 04	0.92	Excellent	Low	Safe	Moderate
SGB 05	1.32	Excellent	Low	Safe	Moderate
SGB 06	0.84	Excellent	Low	Safe	Moderate
SGB 07	0.22	Excellent	Low	Safe	Low
NSB 02	0.21	Excellent	Low	Safe	Moderate
NSB 03	0.19	Excellent	Low	Safe	Moderate
NSB 04	0.18	Excellent	Low	Safe	Moderate

are often location-specific and influenced by well protection, surrounding land use, and hydrogeological vulnerability [54, 83, 87].

4.4.3 Integrated interpretations for borehole prioritization for remediation

The integration of chemical and microbiological data highlights that chemical suitability does not equate to microbiological safety. Boreholes with excellent WQI values but with high microbial counts present a latent health risk if consumed untreated. It is therefore essential to ensure that the safety of drinking water is assessed holistically by using a combined framework that incorporates hydrochemistry, water quality and pollution indices and microbial indicators for a comprehensive risk assessment [22, 89].

Since the chemical suitability of the boreholes was all classified as excellent, the *E. coli* and fecal coliform counts were used to rank the boreholes. Boreholes exhibiting *E. coli* and fecal coliform were ranked as high priority, as these indicators signify recent contamination and an elevated probability of waterborne disease transmission. Boreholes with elevated total coliform counts but no fecal coliform and *E. coli* indicators were classified as moderate priority, reflecting general sanitary deficiencies or surface infiltration pathways rather than direct fecal contamination [90]. In contrast, boreholes with zero detection of microbial indicators and excellent chemical quality were categorized as low priority, requiring routine monitoring rather than immediate intervention (Table 9). This tiered prioritization approach aligns with risk-based groundwater management principles and supports efficient allocation of limited resources, thereby strengthening the interpretation of groundwater beyond conventional chemical indices.

4.5 Groundwater management and policy implications

The finding of the study indicates that although most boreholes exhibit excellent chemical quality based on water quality and pollution indicators, microbial contamination, particularly the detection of *Escherichia coli* and faecal coliforms in JSB 05 and JSB 06, poses a significant public health concern. This divergence between chemical suitability and microbiological safety demonstrates that groundwater sources deemed potable based solely on physicochemical criteria may still present health risks when consumed

untreated. Consequently, groundwater management strategies must integrate chemical, microbiological and risk-based assessment approaches to ensure drinking water safety [15].

4.5.1 Remediation and risk mitigation strategies

Remediation efforts should prioritise boreholes identified as high-risk through integrated chemical-microbial assessment. Structural intervention, including sealing of borehole annuli, construction of sanitary aprons, and improved surface drainage, is essential for limiting contaminant ingress from surrounding sanitation sources. Previous studies in Ghana have demonstrated that inadequate wellhead protection significantly increases susceptibility to faecal contamination in groundwater systems [90].

In addition to structural measures, chemical disinfection techniques such as chlorination and point-of-use treatment remain practical and cost-effective options for reducing microbial loads in rural groundwater supplies [91]. Boreholes located near pit latrines, drainage channels, or animal enclosures were associated with elevated microbial counts, reinforcing the need for targeted remediation rather than uniform interventions. Routine monitoring of microbial indicators (TC, FC, and *E. coli*), supported by WQI evaluation, provides a robust basis for prioritising remediation efforts [54, 83].

The incorporation of Quantitative Microbial Risk Assessment (QMRA) further strengthens remediation planning by translating microbial concentrations into estimated infection risks, thereby supporting evidence-based decision-making regarding intervention urgency and intensity [91, 92].

4.5.2 Preventive strategies and community engagement

Preventive groundwater protection measures are critical to sustaining water safety over the long term. The establishment and enforcement of sanitary protection zones around boreholes, alongside adherence to recommended construction standards, can substantially reduce vulnerability to microbial contamination [22]. Complementary household-level interventions, including hygienic water handling practices and safe storage, further reduce exposure risks [93].

Community engagement formed an integral component of this study, with results disseminated through workshops organised by a collaborating non-governmental organization in partnership with state institutions. These participatory processes enhanced awareness of water quality risks and promoted informed decision-making regarding safe water use. Similar community-based approaches across sub-Saharan Africa have been shown to improve compliance with protective practices and enhance the sustainability of groundwater safety interventions [94, 95].

Low-cost preventive treatment technologies, such as biosand filtration and solar disinfection, may also serve as supplementary measures in high-risk settings where centralized treatment options are limited [96].

4.5.3 Policy implications

The results of this study underscore the need for policy frameworks that explicitly integrate chemical, microbial and risk assessment data in groundwater governance. The adoption of Water Safety Plans (WPS) as advocated by the World Health Organization

offers a structured and proactive approach to hazard identification, risk assessments and implementation of control measures across the water supply chain [90, 91].

Policy instruments should prioritise source protection through enforcement of buffer zones, regulation of on-site sanitation facilities, and routine groundwater quality surveillance. Comparative evidence from Ghana and other West African contexts consistently indicates that microbial contamination remains prevalent even where chemical water quality is acceptable, highlighting the necessity of integrated regulatory oversight [91].

4.5.4 Synthesis

The integrated findings of the study demonstrate that groundwater quality management cannot rely solely on chemical indicators such as WQI and Pollution indicators alone but must also account for microbial contamination and associated public health risks [97, 98]. While most of the boreholes exhibited chemical quality, the presence of *E. coli* and faecal coliform in selected sources confirms that chemically suitable groundwater may still harbour pathogens capable of causing waterborne disease [18, 99].

This evidence highlights the necessity of a multifaceted groundwater management framework wherein remediation strategies such as structural improvements, chlorination and point of use treatment provide immediate risk reduction, while preventive measures, including sanitary protection and community hygiene education, support long-term sustainability [85].

Policy integration further strengthens this framework by enabling coordinated monitoring, early detection of contamination, and prioritisation of high-risk sources through instruments such as Water Safety Plans [90, 91]. By linking hydrochemistry, water quality and pollution indicators, microbiological indicators, and participatory engagement, groundwater management can transition from reactive response to proactive risk prevention.

The synthesis of chemical, microbial, and management evidence provides a robust basis for groundwater governance that is risk-informed, community-supported, and aligned with public health protection and sustainable water resource management objectives in rural and peri-urban environments [100, 101].

5 Limitations of the study

This study was based on sixteen groundwater samples collected during a single sampling campaign, providing a time-specific assessment of groundwater quality. Seasonal variations, particularly between wet and dry periods, may influence hydrochemical concentrations through dilution, evapoconcentration and recharge dynamics. Microbial contamination levels may also vary seasonally due to changes in surface runoff, infiltration pathways and anthropogenic inputs. Furthermore, while major ions and selected microbial indicators were assessed, expanded trace element analysis and long-term monitoring would provide a more comprehensive understanding of groundwater evolution and risk dynamics. Future investigations incorporating multi-seasonal sampling and broader spatial coverage are therefore recommended.

6 Conclusion

This study provides an integrated evaluation of groundwater quality through the combined application of hydrochemical characterization, water quality indices, microbiological assessment and management-oriented interpretation. Hydrogeochemical analysis indicates that ground chemistry in the study area is primarily controlled by natural processes, including mineral dissolution and ion exchange, resulting in water types that are generally suitable for drinking based on physicochemical criteria. Consistently, Water Quality Index results classified all sampled boreholes as excellent for drinking water.

Despite favourable chemical status, microbiological assessment revealed widespread occurrence of total coliforms and localized detection of faecal coliforms and *Escherichia coli* in selected boreholes, indicating sanitary vulnerability and recent fecal contamination at specific locations. This disparity underscores a critical limitation of relying solely on chemical indicators to evaluate groundwater suitability and highlights the necessity of incorporating microbiological and risk-based assessments into drinking water evaluations.

By integrating chemical indices, microbial indicators, and contextual risk interpretation, the study enabled the identification and prioritization of boreholes requiring immediate remediation or enhanced protection. Engagement with local communities, facilitated through collaboration with non-governmental and state institutions, supported the translation of scientific findings into practical guidance on safe water use and exposure risk reduction.

Overall, the findings demonstrate that effective groundwater governance requires an integrated, risk-informed framework that links hydrogeochemical understanding with microbiological safety, public health considerations and participatory management. Such approaches are essential for early detection of contamination, targeted intervention, and the sustainable use of groundwater resources in rural and peri-urban settings [28–31, 38, 39, 69, 71, 74, 78–81, 89, 92, 93, 95, 101].

Supplementary Information

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Supplementary Material 1

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Author contributions

KI conceived the study, developed the research framework, coordinated field sampling, performed data analysis and interpretation, and led the writing of the manuscript. RWK contributed to manuscript drafting and refinement and supported the interpretation of the results. RA and AAY critically reviewed the manuscript and provided technical and scholarly input. MOA and SDS contributed to field support and data validation. All authors reviewed and approved the final manuscript and agree to be accountable for all aspects of the work.

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Data availability

The data used in this study can be obtained from the corresponding author upon request.

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Competing interests

The authors declare no competing interests.

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